Higher-MI Predictor, from First Principles

Can the model's output scores warn us before it suddenly generalises?

Predictor 7 of 8 · *A Unified Benchmark of Grokking Predictors in Neural Networks*

Jonathan John · M.Sc. Artificial Intelligence, IU Internationale Hochschule · 23 October 2026

The result in plain words

A small neural network was taught to add numbers "around a clock" of 113 hours. It first memorised the examples it was trained on. Only much later did it start answering *new* questions correctly. That late, sudden jump is called **grokking**.

We tested one proposed warning signal, called **Higher-MI**. It watches the model's answer scores and asks how much they repeat each other's information. If it worked, its biggest peak would appear *before* the jump.

Verdict: Higher-MI fails as an early warning. Status: CLOSED, negative.

We trained five independent copies of the model (five *seeds*). In all five, the signal's biggest peak came *after* the jump, never before.

Under the project's fixed scoring rules this gives:

- **lead fraction** $= 0$: in 0 of the 5 copies did the peak come early;
- **Spearman** $\rho = 0.7$: copies that jumped later only partly had later peaks (0.9 was required);
- **mean relative lead** $= -0.4254$: on average the peak came about 43% of the jump-time *late*.

Two more facts matter. The signal does not show one clean peak. It rises and falls many times, and its first spike appears at the same epoch, 1,900, in all five copies, even though they jump at very different times. And the result is limited: it shows that *this adaptation*, in *this benchmark*, gives no early warning. It does not show that the underlying idea is useless, nor that the paper it came from is wrong.

The rest of this guide builds every idea needed to understand that verdict, one step at a time. Each step uses only what earlier steps have taught.

Roadmap. A: the task and grokking · B: what a logit is · C: three numbers from 113 logits · D: why three · E: from one example to all test examples · F: information sharing and O-information · G: how it is estimated · H: the curves and the verdict · I: what we learned · Viva questions · Appendix.
Colours: **orange** = grokking; **blue** = the predictor's peak; grey = everything else. Markers and line styles also differ.

Step A. The task, and what grokking is

One example first

The model sees two numbers and must give one answer:

$$(23, 45) \longrightarrow 68.$$

The rule is **modular addition**: $c = (a + b) \bmod 113$. Add the two numbers. If the sum reaches 113 or more, subtract 113. Here $23 + 45 = 68$, which is below 113, so the answer is 68. A second case: $(100, 20) \to 120 - 113 = 7$.

Both inputs lie in $\{0, 1, \ldots, 112\}$. So the answer is always one of exactly **113 possible answers**, 0 to 112. Each possible answer is called a **class**. The model's job is to pick the right class out of 113.

Practice book versus fresh exam

There are $113 \times 113 = 12{,}769$ possible input pairs. They are split randomly into two groups:

- **Training examples** (30%, 3,830 pairs). The model learns from these. This is the practice book.
- **Test examples** (the other 8,939 pairs). The model *never* learns from these. This is the fresh exam paper.

A student who only memorises the practice book scores perfectly on it, yet fails a fresh paper. Test examples tell us whether the model has learned the *rule* or only the *answers*. For seed 0, the pair $(23, 45)$ is a test example; the model never trained on it.

Epochs, seeds and checkpoints

Training runs for 40,000 **epochs**. In each epoch the model sees all 3,830 training pairs once and updates its weights once. A **seed** fixes the random starting weights *and* the random train/test split. Five seeds (0–4) give five independent training runs. Every 100 epochs, the model's weights were saved to disk. Each saved copy is a **checkpoint**: a frozen snapshot of the model at that epoch. Each seed has 401 checkpoints (epochs 0, 100, …, 39,900, plus the final epoch). Everything in this guide was computed later, from these saved snapshots. Nothing was retrained.

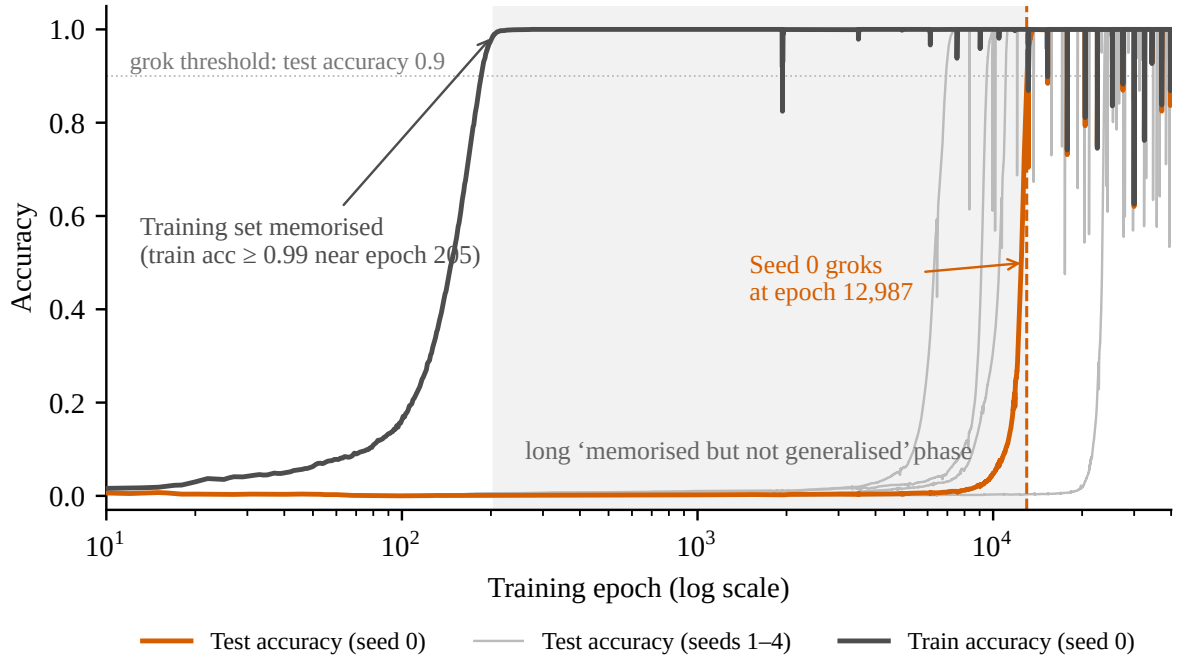
The grokking graph

What do I see? The dark train curve reaches 0.99 near epoch 205 in all five seeds. The orange test curve stays near zero for thousands of epochs after that. Then it rises steeply. For seed 0 it crosses 0.9 at epoch 12,987.

What does it mean? The model memorised the practice book within about 200 epochs, but understood the rule only thousands of epochs later. That delayed, sudden generalisation is **grokking**. We need one number for "when grokking happened". The **grok epoch** G is the first epoch at which test accuracy exceeds 0.9:

Seed	0	1	2	3	4
Grok epoch G	12,987	6,921	9,510	11,019	23,747

Graph 1. When does the model memorise, and when does it generalise?



Saved accuracy histories. Horizontal axis: training epoch, on a logarithmic scale (each tick is 10 times the previous), so that both very early and very late events are visible. Vertical axis: fraction of examples answered correctly. Train accuracy uses the 3,830 training pairs; test accuracy uses the 8,939 unseen pairs.

The same model and settings grok at very different times. This spread is useful. A real warning signal must fire early in *every* seed, and later in seeds that grok later.

The research question. Can a pattern in the model's outputs warn us, *before* G, that grokking is coming? Something that changes only at or after G cannot be a warning.

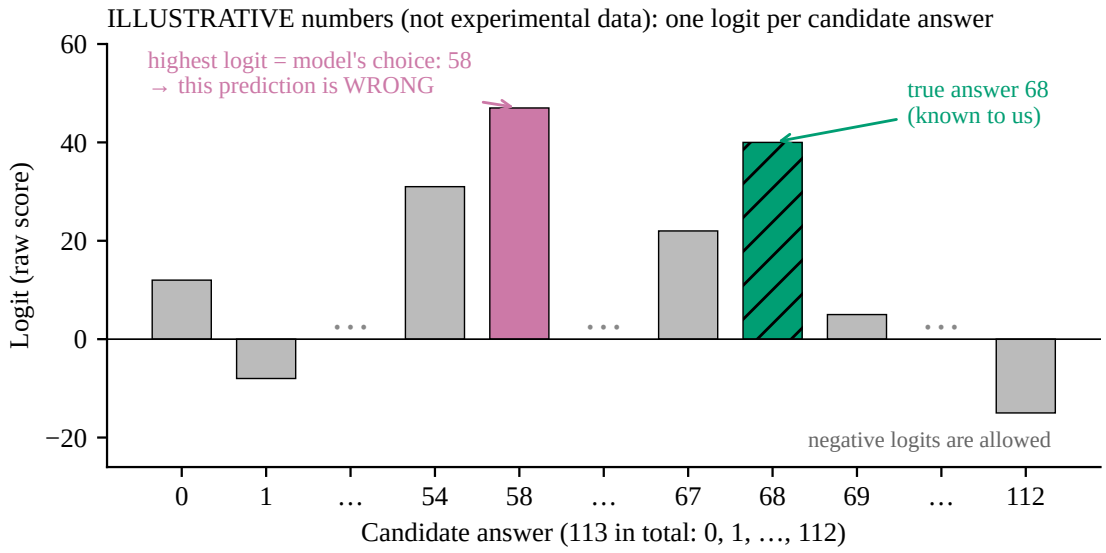
Step B. What the model outputs: logits

For the input $(23, 45)$, the model does not write "68". Instead, it gives **one raw score to every one of the 113 possible answers**. Each raw score is called a **logit**. We write $\ell[j]$ for the logit of answer j.

Five facts about logits:

1. A logit is a **raw, unnormalised score**. Nothing forces the 113 logits to add up to anything.
2. A **larger logit** means the model ranks that answer higher than the others.
3. A logit is **not a probability**. It can be negative, or far larger than 1. (Converting logits into probabilities needs an extra step, softmax, which Higher-MI does not use.)
4. The model's **predicted answer** is the one with the largest logit.
5. The **correct answer** is known to us from the rule. So we can check whether the model's top-ranked answer matches the truth.

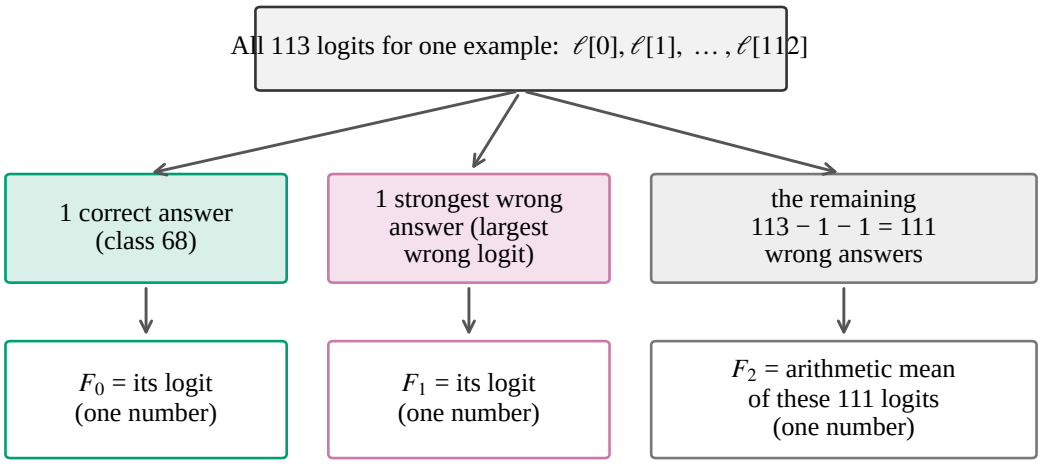
In this model the logits are large, roughly 30 to 150. That is a property of this trained network. It does not affect anything later, because the information measures used below do not change when a variable is rescaled.



Illustrative only. These numbers are invented to show what a logit vector looks like; they are not experimental results.

Step C. Three numbers from 113 logits: F_0, F_1, F_2

Higher-MI does not track all 113 logits. It condenses them into three numbers per example. The figure shows how the 113 answers are divided up.



How the 113 logits of one example become three numbers.

F_0, the correct-answer logit. The logit of the answer that is actually correct. For $(23,45)$ this is $F_0 = \ell[68]$.

F_1, the strongest wrong-answer logit. Look at the 112 wrong answers. Take the largest logit among them: $F_1 = \max_{j \neq 68} \ell[j]$. The wrong answer that achieves it is the **runner-up**, r. It is the most dangerous competitor.

F_2, the background logit. Start with all 113 answers. Remove the correct answer (one). Remove the runner-up (one more). Exactly $113 - 1 - 1 = 111$ wrong answers *remain*. F_2 is the plain arithmetic

average of their 111 logits:

$$F_2 = \frac{1}{111} \sum_{j \neq c,\, j \neq r} \ell[j].$$

F_2 is nothing more than that average. It is not a probability, a confidence or an uncertainty.

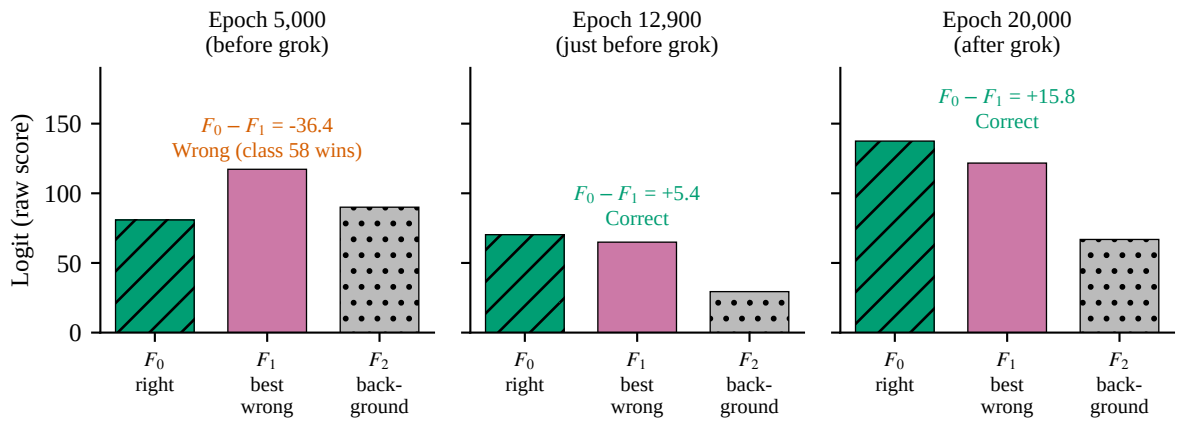
When is the model right? The model picks the largest logit. So it is right exactly when the correct answer beats every wrong answer, which means beating the strongest one:

$$\text{correct} \iff F_0 > F_1.$$

The difference $F_0 - F_1$ is the **margin**. Positive margin: correct. Negative margin: wrong. A large positive margin: a decisive, confident win.

The real example at three checkpoints

Graph 4. One unseen example, (23, 45) → 68, seed 0: how the three scores change



Real values for $(23, 45) \to 68$, seed 0, computed from the saved checkpoints. Bar height is the logit. The text above each panel is the margin $F_0 - F_1$.

What do I see? At epoch 5,000 the wrong answer 58 beats the correct answer 68 by about 36. Even the background average F_2 is above F_0. At epoch 12,900, just before seed 0 groks, 68 has moved to the top (margin +5.4; the runner-up is now 54) and the background has fallen far below. At epoch 20,000, after grokking, the margin has grown to about +15.8.

What does it mean? At epoch 5,000 the model had already memorised the training pairs, but on this unseen pair it had not learned the rule. Later it had. This is *one* example. It shows the mechanics of F_0, F_1 and F_2. It proves nothing about the model as a whole.

Step D. Why exactly these three numbers?

Higher-MI comes from Pomarico *et al.* (2025, arXiv:2507.23346). They studied a different model (a matrix product state classifier, from tensor-network physics) on **three-class** problems: a reduced Fashion-MNIST (dress, sneaker, bag) and hyperspectral land-cover data. With three classes, the model naturally produces exactly three scores, and their information measure (Step F) is built for three variables.

Our task has 113 classes. So the project needed some rule to turn 113 logits into three variables. It keeps

the three quantities that decide each prediction: the score to beat (F_0), the strongest challenger (F_1) and the level of everyone else (F_2). This **reduction is the project's own adaptation**. The paper does not prescribe it.

What is lost. The reduction keeps only one challenger and one average. It forgets *which* wrong answer is the runner-up, how the 111 remaining logits are spread around their mean, and whether several wrong answers are close to F_1. A different reduction could give a different signal. This is recorded as an assumption.

The "=" output. The model reads each pair as three tokens: "*a b* =". Its output layer therefore has 114 entries: the 113 numbers plus one for the "=" token. The "=" entry is never a correct answer, so the model is never trained to make it high or low. If kept, it could become a false runner-up or shift F_2. It is removed before F_0, F_1 and F_2 are computed.

Step E. From one example to the whole test set

So far: one example gives three numbers. Now fix one checkpoint and do this for *every* test example.

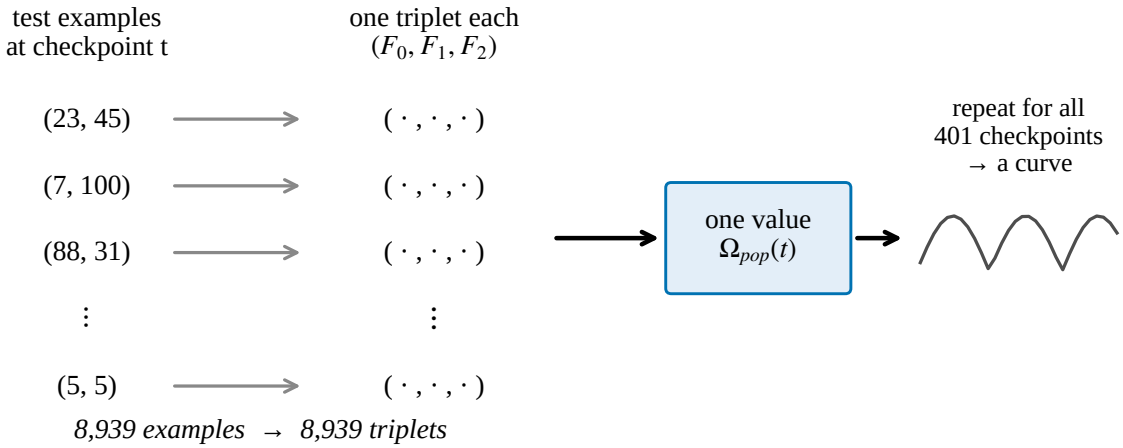


Diagram. The listed input pairs are placeholders showing the procedure.

- One example → three numbers (F_0, F_1, F_2).
- All 8,939 test examples, at one checkpoint → 8,939 triplets.
- Those 8,939 triplets → **one** information value for that checkpoint (Step F).
- Repeat at all 401 checkpoints → one **curve** per seed.

Why do the triplets differ? Some pairs are easy for the model at that moment: high F_0, lower F_1, very low F_2. Others are confusing: F_0 and F_1 close together. So across the test set, F_0, F_1 and F_2 behave like three **random variables**: each example is one draw of the three together. (The fraction of triplets with $F_0 > F_1$ is exactly the test accuracy.)

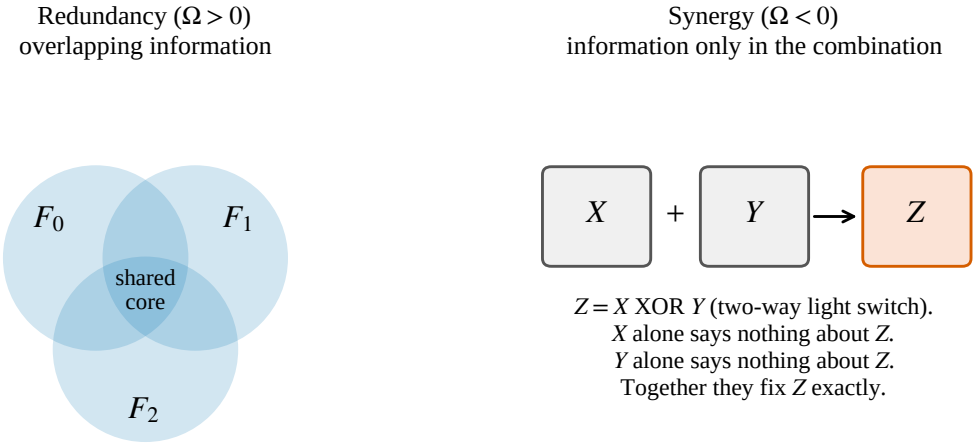
Why test examples? Grokking is a change in how the model handles *unseen* inputs. Training pairs are memorised early and confidently, so they would hide that change. Using test examples is the right place to look. It is not, by itself, proof that any signal found there predicts anything.

Step F. Information sharing, then O-information

Two ways three variables can relate

Redundancy: the same news, repeated. Three newspapers report the same cricket score. Read one and the other two add nothing. Their information *overlaps*.

Synergy: meaning only in combination. A staircase light has two switches, one downstairs and one upstairs. The bulb is on when the switches are in different positions. Look at one switch: it tells you nothing about the bulb. Look at both: you know the bulb exactly. The information exists only *jointly*. (This is the XOR rule.)



Schematic. Left: overlapping information. Right: information present only in the combination.

Measuring shared information

Mutual information, $I(X;Y)$, answers: *how much does knowing X reduce my uncertainty about Y?* If X and Y are unrelated, it is 0. The more one pins down the other, the larger it is. It is measured in **nats** (information units based on the natural logarithm). It captures any kind of dependence, not only straight-line correlation.

Conditional mutual information, $I(X;Y \mid Z)$, answers: *once I already know Z, how much does X still tell me about Y?*

Now compare the two for our three variables. If F_2 already explains much of what links F_0 and F_1, then knowing F_2 *shrinks* their shared information. That is overlap, i.e. redundancy. If knowing F_2 *increases* what F_0 and F_1 share, the link appears only in combination. That is synergy.

O-information

O-information turns this comparison into one number:

$$\Omega(F_0,F_1,F_2) = \underbrace{I(F_0;F_1)}_{\text{what correct and runner-up share}} - \underbrace{I(F_0;F_1 \mid F_2)}_{\text{what they still share once the background is known}} .$$

- $\Omega > 0$: redundancy dominates. The three logit summaries carry overlapping information.

- $\Omega < 0$: synergy dominates. Some information exists only in the triple.

This is the sign convention of the source paper, and it is kept here.

What the sign does not say. Ω describes the statistical relationship among the three numbers across examples. It does not directly say whether the model has grokked. A positive value is not "grokked"; a negative value is not "not grokked". Also, large shared information does not imply large Ω. At epoch 20,000, F_0 and F_1 move almost in lockstep across examples (correlation 0.998), yet Ω is only about 0.36 nats, because Ω counts only the part of the link that F_2 explains away.

Population O-information. When Ω is computed from the 8,939 test-example triplets at one checkpoint t, we call it **population O-information**, $\Omega_{\text{pop}}(t)$. It describes structure *across examples at one frozen moment*. It is *not* a measure of how the three scores move together over training time. (Step H returns to this difference.)

The hypothesis. The paper reports strongly negative O-information at initialisation, a switch to positive as the model learns, and a redundancy peak *that coincides with* grokking. This thesis tests a stronger claim:

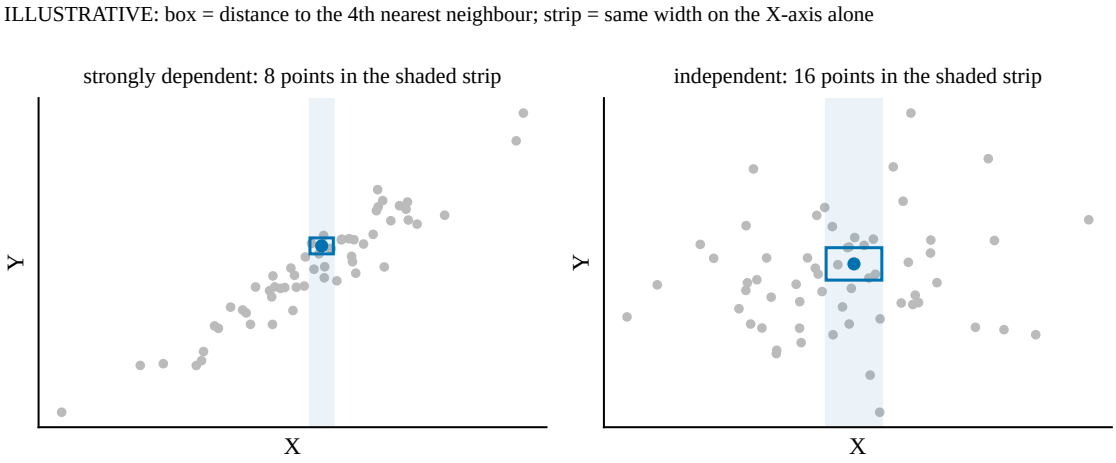
> *The peak of $\Omega_{\text{pop}}(t)$ comes before the grok epoch, with timing that follows the grok epoch across seeds.*

Step G. How Ω_{pop} is estimated

The problem. Mutual information is defined from probability distributions. We do not know the distributions of F_0, F_1, F_2; we only have 8,939 sample points.

The obvious fix, and its flaw. Cut each axis into bins and count points per bin. But the answer changes with the bin width, and the width is an arbitrary choice.

The KSG estimator (Kraskov, Stögbauer and Grassberger) avoids bins by using distances between points.



Illustrative synthetic points, not experimental data. Both panels use the same rule.

For each point, draw the smallest square box that reaches its 4th nearest neighbour. Then ask: on the X-axis *alone*, how many points fall inside the same width (the shaded strip)? If X and Y are strongly

dependent, points cluster along a line, so the box is small and few points share that narrow strip. If they are independent, many points fall in the strip anyway. Few strip neighbours relative to the box means high mutual information. KSG converts these counts into a number.

Settings: $k = 4$ neighbours. In theory Ω treats its three variables symmetrically, but the estimate does not exactly, so Ω is computed for all 6 orderings of (F_0, F_1, F_2) and averaged. The formulas are in the Appendix.

Is the estimator trustworthy? It passes known-answer tests: unrelated variables give $I \approx 0$; near-copies give $\Omega > 0$; an XOR-like product construction gives $\Omega < 0$; a small fake model with hand-chosen logits gives the right (F_0, F_1, F_2) and ignores "=". These checks show that the *measurement is correct*. They do not show that the measured signal *predicts grokking*. That is tested next.

Step H. The curves and the verdict

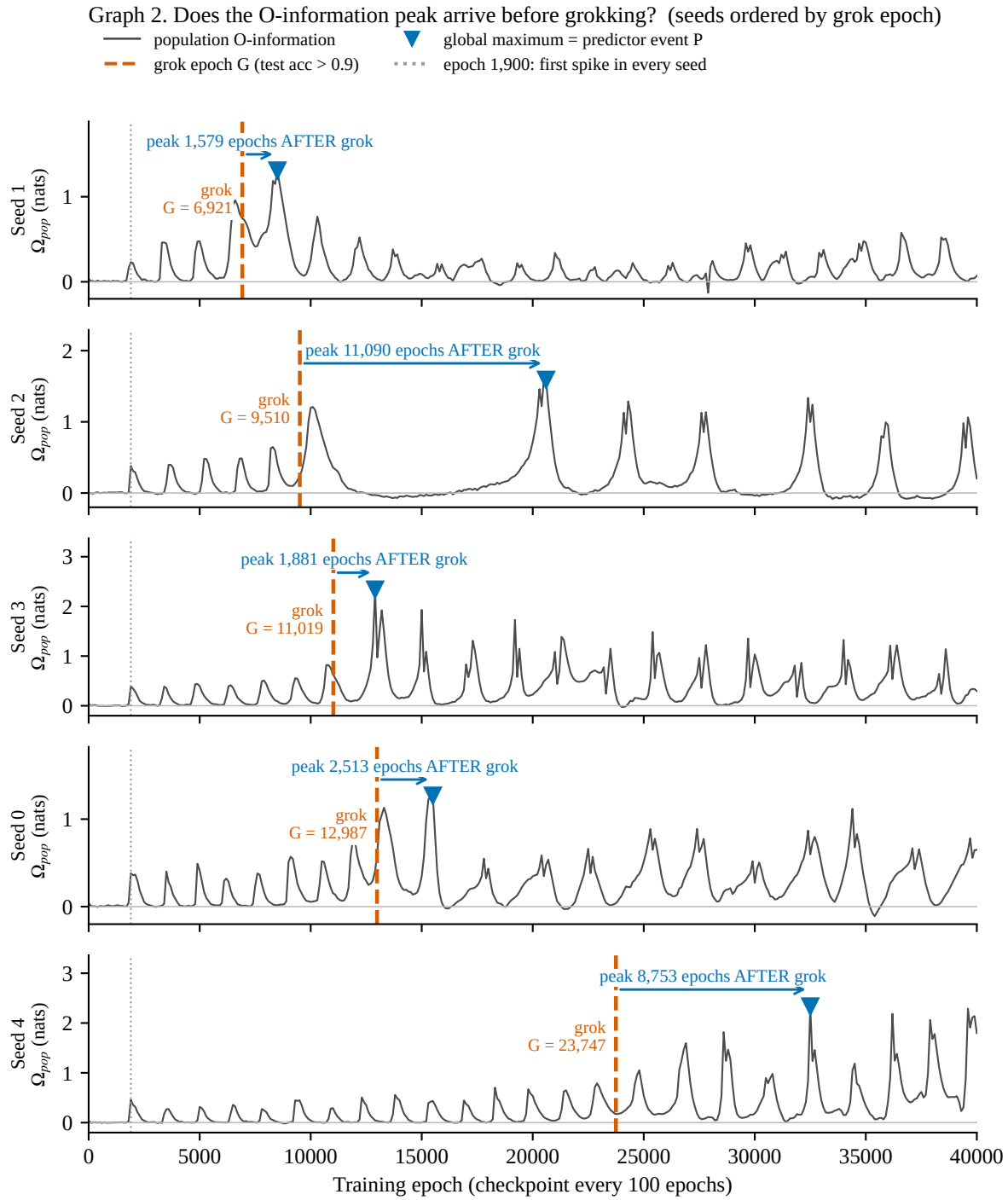
From a curve to one event epoch

For each seed we now have $\Omega_{\text{pop}}(t)$ at 401 checkpoints. **Upward** movement means the three logit summaries became more redundant across test examples. **Downward** movement towards 0 means less redundancy; below 0, synergy. To compare with G, each curve must be reduced to one epoch, by a rule fixed *before* looking at the score. The rule, frozen on 3 October 2026, is the **global maximum**: $P =$ the checkpoint where Ω_{pop} is highest. This is the paper's own landmark (the redundancy peak), it always exists, and it matches the other predictors in the thesis.

What do I see?

- **No strongly negative start.** At epoch 0, Ω_{pop} lies between $+0.002$ and $+0.040$. Negative values are rare and small (lowest -0.130).
- **Repeated spikes, not one transition.** The curve shoots up and falls back many times, before and after grokking. A grokking warning should show one change tied to G; a repeating pattern has no single moment to point at.
- **A common clock.** In all five seeds, Ω_{pop} first exceeds 0.2 at exactly epoch 1,900, then (before grokking) spikes roughly every 1,400–1,600 epochs.
- **No mark at grok.** The orange line falls in the middle of the oscillation.
- **The highest spike is always to the right of the orange line.**
- **Different growth per seed.** Seed 4 rises from about 0.04 to 1.80; seed 1 ends near where it started (0.033 to 0.070).

What does it mean? The seeds grok between epochs 6,921 and 23,747, but the spikes start at epoch 1,900 in every seed. A pattern on the same clock for every seed cannot be specific to grokking. It more likely follows general training dynamics, such as how margins develop. That is an interpretation, not a tested fact; the cause of the epoch-1,900 onset was not investigated and remains open. The paper's sign story (strongly negative, then positive) is not reproduced here.



Saved $\Omega_{\text{pop}}(t)$ for all five seeds, all 401 checkpoints, no smoothing. Each panel uses its own vertical range. Orange dashed line: grok epoch G. Blue triangle: global maximum P. Grey dotted line: epoch 1,900.

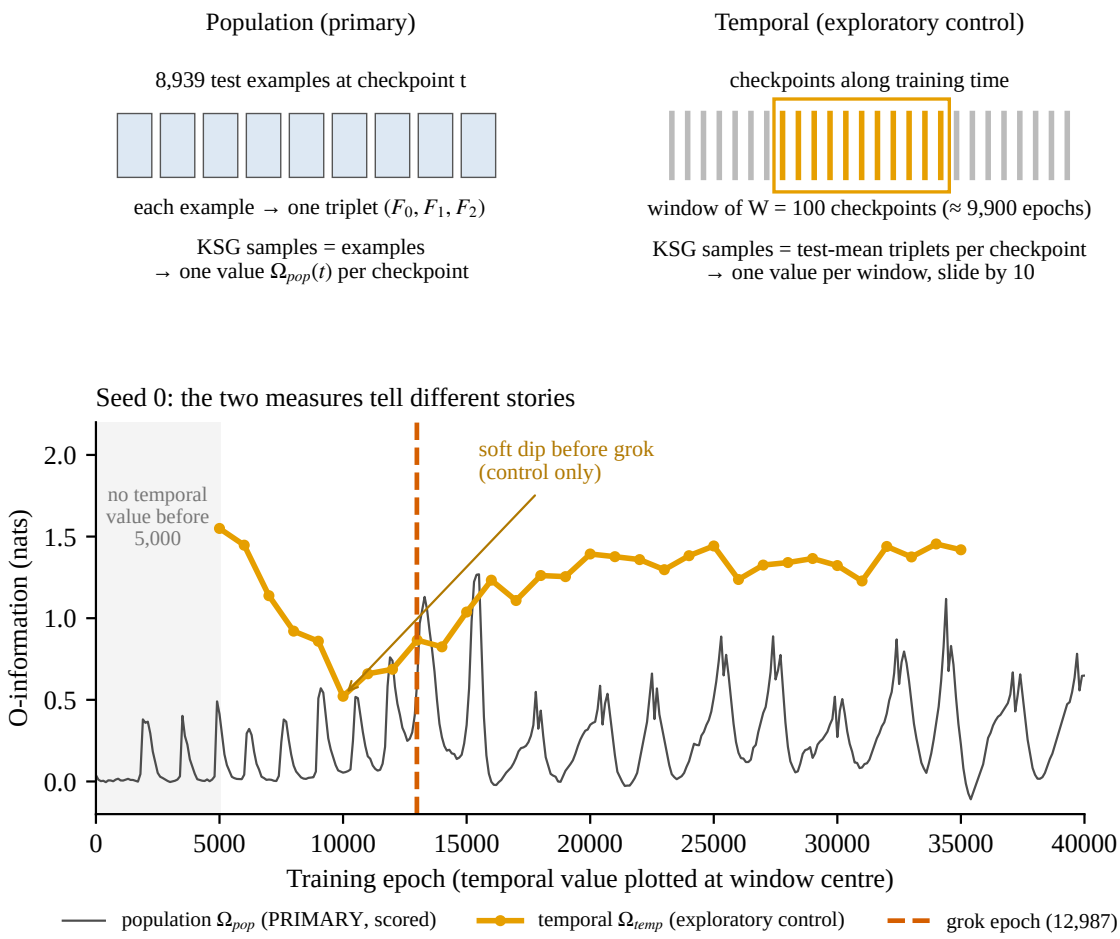
Population versus temporal O-information

A worry: shared example difficulty. F_0, F_1 and F_2 come from one forward pass on the same input. Easy examples tend to be high-F_0/low-F_2 together; hard ones have F_0 near F_1. That common cause, difficulty, can create redundancy by itself, grokked or not.

A control, in the paper's own sense of "sample". In the paper, one *sample* is one training step: O-information there describes how the three scores move together over time. In Ω_{pop}, one sample is one

test example at a fixed checkpoint. The exploratory **temporal control** Ω_{temp} follows the paper: average F_0, F_1, F_2 over the test set at each checkpoint, then treat a window of 100 consecutive checkpoints (about 9,900 epochs) as the samples. The window slides by 10 checkpoints, giving 31 values, centred from epoch 5,000 to 35,000.

Graph 5. Population versus temporal O-information



Top: what one sample is in each measure. Bottom: both saved curves for seed 0. Each temporal value is plotted at its window centre.

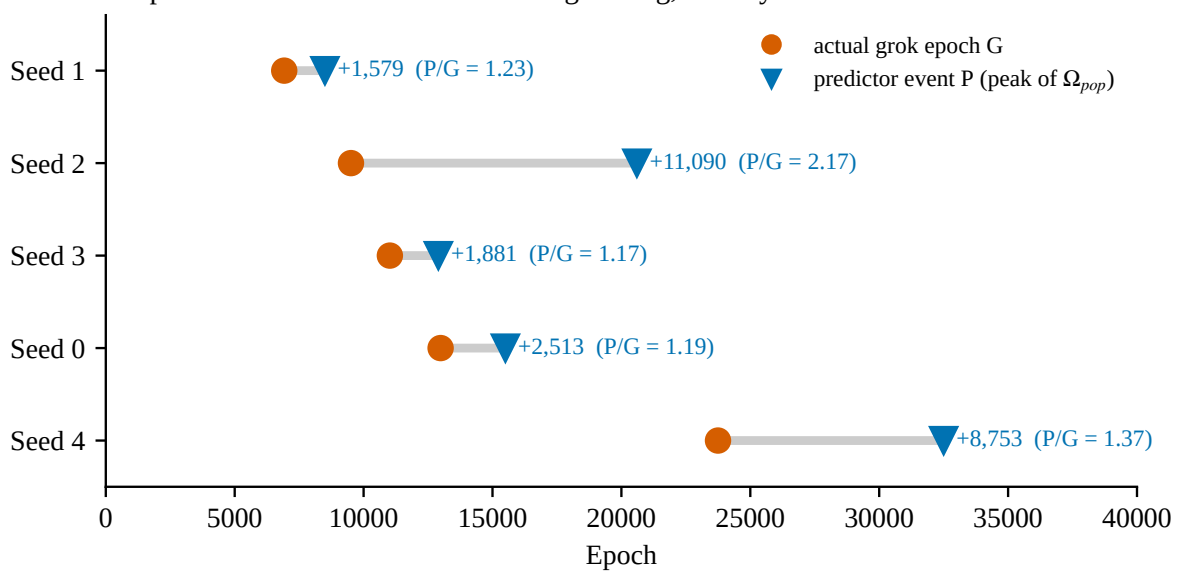
What do I see? The temporal curve is smooth. It starts near 1.55, dips to about 0.52 near epoch 10,000, just before grok, then stays near 1.2–1.45, never negative in any seed. The population curve shows no such dip.

What does it mean? The two shapes differ, so Ω_{pop} is not simply a copy of one difficulty factor. This partly eases the confound worry; it does not remove it. The two measures are *not interchangeable*. The soft dip belongs to the unscored control, whose wide windows cannot time it precisely; it does not rescue the scored predictor.

Timing: warning, coincidence or late?

An early warning puts the triangle *left* of the circle. A coincident signal puts them on top of each other. A late signal puts the triangle to the right.

Graph 3. Predicted event versus actual grokking, seed by seed



Grok epoch G (orange circle) and peak epoch P (blue triangle) per seed. $P/G > 1$ means the peak came after grokking.

What do I see? All five triangles are to the right, at 1.17 to 2.17 times G. Seed 1 is earliest in both G and P, and seed 4 latest in both, but seeds 2 and 3 swap.

What does it mean? Higher-MI is a late signal in every seed. It partly follows the order of grokking, but never comes first.

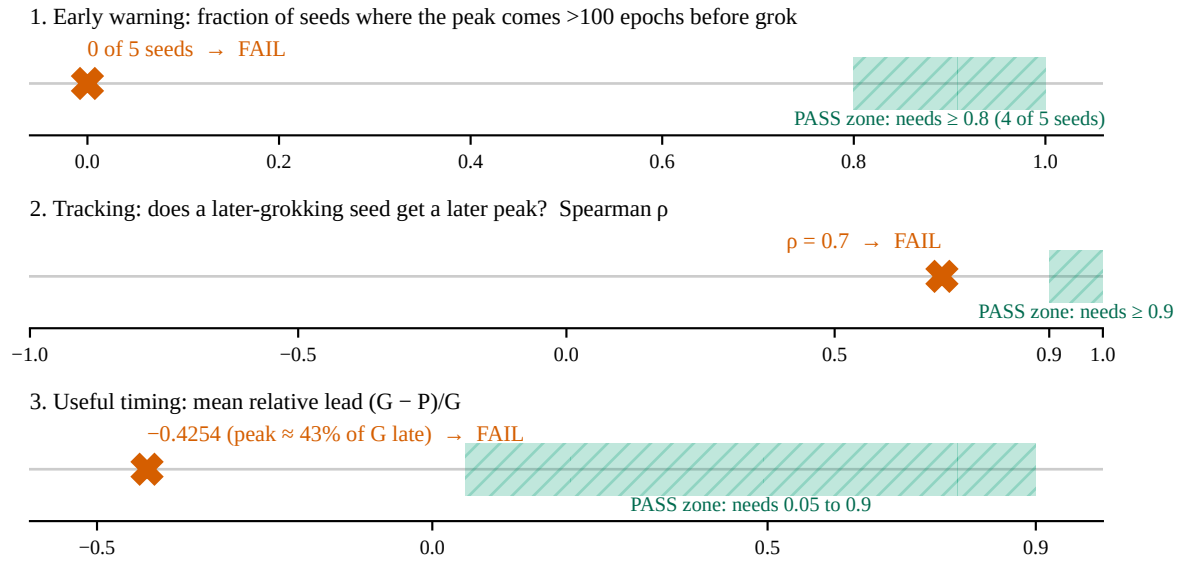
The frozen scoring rule

The same rule is used for every predictor in the thesis, and all three conditions must pass:

1. **Lead fraction** ≥ 0.8**.** A seed "leads" if P is more than 100 epochs before G. At least 4 of 5 seeds must lead.
2. **Spearman** $\rho \geq 0.9$**.** Rank the seeds by G and by P; ρ measures how well the two orders agree (1 = identical order). The 0.9 bar was calibrated: random predictors exceed it only 1% of the time.
3. **Mean relative lead between 0.05 and 0.9.** For each seed, $(G-P)/G$: how early the peak is, as a fraction of that seed's grok time. Negative means late. Below 0.05 is too late to use; above 0.9 fires almost at the start and carries no timing information.

Practical meaning. Lead fraction 0: no seed gave any warning. $\rho = 0.7$: the peak order only loosely follows the grok order. Mean relative lead -0.4254: the peak arrived, on average, about 43% of a grok-time *after* grokking. All three fail, so the verdict is FAIL. It is not borderline.

Graph 6. The frozen-protocol scorecard: all three conditions must pass



Hatched green band: pass zone. Cross: Higher-MI's value.

Step I. What we learned

Observed. Ω_{pop} peaks after grokking in 5 of 5 seeds, oscillates throughout training, and starts spiking at epoch 1,900 in every seed. **Interpreted.** It appears to track general training dynamics rather than the grokking transition. **Status.** Higher-MI is closed as negative: the seventh predictor to do so, after L2 Norm, Dropout, Spectral, AGE, HTSR Alpha and Weight-PCA. Commutator Defect is next.

This does not contradict the paper

	Pomarico *et al.* (2025)	**This project**
Model	Matrix product state classifier	One-layer transformer
Task	Three-class problems	Modular addition, 113 classes
Variables	The three class scores	Project-built (F_0, F_1, F_2)
One sample	One training step (temporal)	One test example (population)
Claim	Peak *coincides* with grokking	Peak *leads* grokking

The paper never claimed a lead. Here the peak is not even coincident; but model, task, variables and sampling all differ, so this cannot refute the paper in its own setting.

Why the numbers can be trusted

Two silent bugs were found and fixed before scoring:

- **Inclusive versus strict counting.** KSG must count points *strictly* inside each distance. The library call counted "less than or equal", which wrongly includes the very neighbour that sets the distance. About half the points were affected in one cross-check; one test case changed Ω about 2.6-fold. Fixed by querying just below the distance; it then matched a brute-force check bit for bit.
- **Test split not re-seeded.** The recompute code rebuilt the test set without first setting the seed, so

many "test" pairs were really training pairs. This would have biased exactly the early checkpoints a warning needs. Fixed by rebuilding the split with the saved seed. The rebuilt seed-0 split has 8,939 pairs and reproduces the stored Ω_{pop} values exactly.

The scoring rule passes an ideal synthetic predictor 100% of the time. So the pipeline can recognise a working predictor; Higher-MI simply is not one here.

Assumptions, limitations and open questions

- **Assumptions:** the (F_0, F_1, F_2) reduction; samples = test examples; $k = 4$ (the paper states it only for its transfer-entropy analysis); dropping "="; the global-maximum rule.
- **Only five seeds.** ρ can take only 11 values. A random predictor that always fires before grok passes the rule about 22% of the time. Higher-MI fails on lead, so this does not change its verdict.
- **100-epoch checkpoint spacing.** Faster features are invisible.
- **Coarse temporal control.** Each window spans about 9,900 epochs; no value before epoch 5,000.
- **Shared-difficulty confound.** Partly addressed, not removed.
- **The epoch-1,900 onset.** Unexplained. This question is still open.

Further work would need more seeds, denser checkpoints near epoch 1,900 and near each G, an explanation of the common onset, a control for example difficulty, and other reductions of the 113 logits.

What I must be able to explain

1. **What is grokking?** Delayed generalisation: train accuracy is near perfect by epoch $\sim$205, but accuracy on unseen pairs crosses 0.9 only thousands of epochs later (6,921–23,747 here).
2. **What is a logit?** The raw, unnormalised score the model gives each of the 113 answers. Higher means ranked higher; it is not a probability; the largest one is the prediction.
3. **What are F_0, F_1, F_2?** The correct answer's logit; the largest wrong-answer logit; the average logit of the 111 wrong answers left after removing the correct answer and that runner-up. The model is right exactly when $F_0 > F_1$.
4. **Why three numbers?** O-information as used is a three-variable quantity; the paper had three classes. Reducing 113 logits to three is our adaptation, and it discards detail.
5. **Why unseen test examples?** Grokking is a change on unseen inputs. Training pairs are memorised early and hide it. Mixing them in was Bug 2.
6. **Redundancy and synergy?** Overlapping information (repeated newspapers) versus information present only jointly (two-switch light). $\Omega > 0$ redundancy, $\Omega < 0$ synergy.
7. **What is O-information here?** $I(F_0;F_1) - I(F_0;F_1 \mid F_2)$: how much of the link between correct and runner-up logits is explained away by the background average, estimated across the 8,939 test examples at one checkpoint.
8. **Population versus temporal?** Population: samples are test examples at one checkpoint. Temporal: samples are checkpoints in a time window. Different questions; different curves.
9. **Why the global maximum?** The paper's landmark, always defined, consistent with the other predictors, frozen before scoring.

10. **Why did it fail?** Peak after grokking in all five seeds: lead fraction 0, $\rho = 0.7$, mean relative lead -0.4254. All three conditions fail.

11. **Is the paper wrong?** No. Different model, task, variables and sampling; the paper claimed coincidence, we tested lead.

12. **Limitations?** Five seeds, 100-epoch spacing, coarse temporal control, difficulty confound, unexplained epoch-1,900 onset.

Appendix: technical details

A1. KSG formulas. Distances use the maximum norm. For point i, ε_i is the distance to its k-th nearest other point in the joint space; $N_x(i) = \#\{j \neq i : |x_j - x_i| < \varepsilon_i\}$ (strict). With ψ the digamma function, T samples and $\langle\cdot\rangle$ the average over points:

$$I(X;Y) = \psi(k) + \psi(T) - \langle \psi(N_x+1) + \psi(N_y+1) \rangle,$$

$$I(X;Y \mid Z) = \psi(k) + \langle \psi(N_z+1) - \psi(N_{xz}+1) - \psi(N_{yz}+1) \rangle.$$

$\Omega_{\text{pop}}(t)$ is the mean of $I(F_{\pi_1};F_{\pi_2}) - I(F_{\pi_1};F_{\pi_2} \mid F_{\pi_3})$ over the 6 orderings π. At seed 0, epoch 12,900: mean 0.4044, standard deviation 0.0099. The code computes $F_2 = (\sum_j \ell[j] - F_0 - F_1)/111$.

A2. Protocol. With margin 100: lead_frac $= \frac{1}{5}\sum_s \mathbf{1}[P_s < G_s - 100]$, mean_rel_lead $= \frac{1}{5}\sum_s (G_s - P_s)/G_s$, $\rho =$ Spearman(G,P).

Seed	0	1	2	3	4
G_s	12,987	6,921	9,510	11,019	23,747
P_s	15,500	8,500	20,600	12,900	32,500
$\Omega_{\text{pop}}(P_s)$	1.2689	1.3089	1.5917	2.3403	2.3347
$(G_s - P_s)/G_s$	−0.1935	−0.2281	−1.1661	−0.1707	−0.3686

Ranks of G: 4,1,2,3,5; of P: 3,1,4,2,5; $\sum d^2 = 6$; $\rho = 1 - 36/120 = 0.7$ (Pearson 0.8976; the rule uses rank correlation). The 0.9 threshold is the 99th percentile of ρ over 10,000 random predictors $P \sim U[0,40000]$. Synthetic controls, pass rates: ideal $P = 0.5G$ 100%; constant $P = 5000$ 0%; $P = 1.5G$ 0%; noisy ideal 91.04%; random 0.22%; random but always before grok 21.70%.

A3. Exploratory signals (not scored). Population zero-crossing: $P_s = 500, 1600, 300, 1100, 200$; lead_frac 1.0, $\rho = -0.7$, mean_rel_lead 0.9181: noise near initialisation, the same failure as the constant-early control. Temporal-control maximum: $P_s = 5000, 5000, 19000, 5000, 14000$; lead_frac 0.8, mean_rel_lead 0.1703; three "leads" are just the first window centre. *Flagged discrepancy:* stored $\rho = 0.00$ (position-based tie-breaking) versus tie-aware 0.11; both far below 0.9. Transfer entropy $TE_{X\to Y}(\tau) = I(Y(t+\tau);X(t) \mid Y(t))$ between test-mean correct and runner-up logits, $\tau = 1..10$: at $\tau = 1$ correct→runner-up exceeds the reverse in all seeds; on strongly trending trajectories, so not a causal result.

A4. Model and data. One-layer transformer, 4 heads of width 32, $d_{\text{model}} = 128$, MLP 512 (ReLU), no LayerNorm, learned positions, untied embeddings; AdamW, learning rate 10^{-3}, weight decay 1.0, $\beta = (0.9, 0.98)$, full batch, 40,000 epochs. The shared model file and training loop were not changed for Higher-MI.

A5. Sources and reproduction. Curves and scores are in `08_Experiments/results/nanda_unified/`. Files: `seed_<N>/higher_mi/*`; `aggregate_higher_mi.json`; `seed_<N>/training/*_acc_history.npy`. Graph of $(23, 45)$ logits: values from the source explanation. Full recompute ≈27 min:

```
python 06_Code/scripts/run_nanda_benchmark.py \
    --predictors higher_mi --seeds 5 --overwrite higher_mi \
    --output_dir 08_Experiments/results/nanda_unified
python 06_Code/scripts/analyze_nanda_unified.py
```

Rounding: the thesis wording quotes −0.43; the stored value is −0.4254.